

■ Bems Farms — Driver Mobile App API Specification

Official Engineering Guide for Mobile App Developers (React Native, Flutter, iOS & Android) • Version 2.0.0

Target Mobile Platforms: React Native / Flutter / iOS / Android	Authentication: JWT Bearer (`Authorization: Bearer <token>`)
Production API URL: https://api.bemsfarms.com/api/driver	Local Development URL: http://localhost:5000/api/driver
Payload Standard: application/json & multipart/form-data	Driver Wallet & DVA: Monnify Virtual NUBAN + Zone-Based Earnings

1. Driver API Directory & Endpoint Index (24 Endpoints)

Category	Method & Endpoint	Auth	Description & Mobile Trigger
Onboarding	GET /onboarding/verify?token=...	Public	Verify invite token & pre-populate driver signup details
Onboarding	POST /onboarding/submit	Public	Submit driver KYC, NIN, vehicle details & settlement bank
Auth	POST /auth/login	Public	Driver email/phone + password authentication & JWT token
Auth	POST /auth/forgot-password	Public	Send password reset token via SMS / Email
Auth	POST /auth/reset-password	Public	Confirm reset token and configure new driver password
Profile	GET /auth/me	Bearer	Retrieve driver profile, rating, stats & Monnify Virtual Account
Profile	PATCH /auth/profile	Bearer	Update driver phone number, profile photo, or vehicle details
Availability	PATCH /availability	Bearer	Toggle duty state: Online (ready for dispatch) vs Offline
Deliveries	GET /deliveries	Bearer	Fetch assigned active deliveries with customer addresses & items
Deliveries	GET /deliveries/:orderId	Bearer	Fetch full itemized breakdown for a specific delivery drop
Deliveries	POST /deliveries/:orderId/accept	Bearer	Explicitly accept an assigned delivery drop within countdown
Deliveries	POST /deliveries/:orderId/decline	Bearer	Decline delivery with reason so dispatch can reassign
Milestones	PATCH /deliveries/:orderId/status	Bearer	Progress milestone (accepted, en_route, arrived, delivered)
Incidents	POST /deliveries/:orderId/report-issue	Bearer	Report mid-trip issue (customer unreachable, wrong address)
Incidents	GET /incidents	Bearer	List past incident reports and resolution statuses
History	GET /deliveries/history	Bearer	Fetch paginated past delivery drops and completed trips
POD Upload	POST /upload/proof	Bearer	Upload camera photo proof of delivery (multipart/form-data)
Telemetry	POST /location	Bearer	Background GPS coordinate stream ping (every 10–15s)
Wallet	GET /earnings	Bearer	Live wallet balance, Monnify DVA NUBAN, zone rate card & history
Bank Directory	GET /banks	Bearer	Directory of Nigerian commercial banks & fintechs with codes
Bank Resolve	POST /bank/resolve	Bearer	Verify 10-digit NUBAN account number & resolve legal account name
Withdrawal	POST /withdraw	Bearer	Request payout of unwithdrawn earnings to personal bank account
Push Tokens	POST /device-token	Bearer	Register / update device Expo / FCM push notification token
Notifications	GET /notifications	Bearer	Fetch in-app notification inbox with unread count
Notifications	PATCH /notifications/:id/read	Bearer	Mark single notification as read
Notifications	PATCH /notifications/read-all	Bearer	Mark all notifications in inbox as read
Performance	GET /stats /performance	Bearer	Driver scorecard, rating (■), on-time rate & badges
Emergency SOS	POST /emergency	Bearer	■ 1-Tap SOS panic alert broadcasting live GPS to dispatch
Emergency SOS	POST /emergency/:id/cancel	Bearer	Cancel emergency false alarm

2. Driver Authentication, Onboarding & Profile APIs

2.1 Driver Login (`POST /api/driver/auth/login`)

Authenticates driver by phone number or email and password. Returns JWT token and driver profile with Dedicated Virtual Account (DVA).

Request Payload	Success Response (200 OK)
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<pre>{ "emailOrPhone": "08123456789", "password": "Password123!" }</pre>	<pre>{ "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9... ", "message": "Login successful", "driver": { "id": 6, "name": "Victor Kalu", "phone": "08123456789", "email": "victor.kalu@bemsfarms.com", "wallet_account_number": "3397202109", "wallet_bank_name": "Monnify / Wema Bank", "wallet_account_name": "BEMS - VICTOR KALU", "bank_name": "GTBank", "account_number": "0123456789", "wallet_balance": 16500.00, "is_available": true } }</pre>
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2.2 Toggle Availability State (`PATCH /api/driver/availability`)

Request Payload	Success Response (200 OK)
<pre>{ "is_available": true, "latitude": 9.0765, "longitude": 7.3986 }</pre>	<pre>{ "message": "Duty status updated to Online", "is_available": true, "status": "active" }</pre>

3. Dispatch Deliveries, Accept/Decline & Milestones

When an order is dispatched, the driver has endpoints to explicitly Accept or Decline, progress through 5 status milestones, and capture photo proof of delivery.

3.1 Accept Delivery Drop (`POST /api/driver/deliveries/:orderId/accept`)

<pre>{ "status": "success", "message": "Delivery accepted successfully", "delivery_id": 482, "order_id": 1092, "order_ref": "BEMS-ORD-8819", "status": "accepted" }</pre>

3.2 Decline Delivery Drop (`POST /api/driver/deliveries/:orderId/decline`)

Request Payload	Success Response (200 OK)
<pre>{ "reason": "vehicle_breakdown", "notes": "Flat tyre along Airport Road, fixing at repair shop." }</pre>	<pre>{ "status": "success", "message": "Delivery declined. Returned to dispatch pool for reassignment.", "order_id": 1092, "order_ref": "BEMS-ORD-8819" }</pre>

3.3 Milestone Progression Lifecycle

Milestone	Action Description	Customer Tracking State	Financial Trigger
accepted	Driver accepts delivery assignment	driver_assigned	—
awaiting_pickup	Driver at hub/store loading goods	packed_ready	—
en_route	Driver departed with order package	out_for_delivery	—
arrived	Driver arrived at customer destination	driver_arrived	—
delivered	Handover complete with photo proof	delivered	Auto-Credit Zone Commission
failed	Delivery attempted but customer unavailable	delivery_attempted	—

3.4 Update Milestone (`PATCH /api/driver/deliveries/:orderId/status`)

Request Payload	Success Response (200 OK)
<pre>{ "status": "delivered", "proof_note": "Handed directly to customer Mrs. Ngozi.", "proof_photo": "https://api.bemsfarms.com/uploads/proofs/POD_17899_a8b9.jpg" }</pre>	<pre>{ "message": "Delivery status updated to delivered", "delivery": { "id": 482, "status": "delivered", "arrived_at": "2026-09-21T10:14:00Z" }, "order_id": 1092, "status": "delivered" }</pre>

4. Proof of Delivery (POD) Photo Upload API

4.1 Upload POD Photo (POST /api/driver/upload/proof)

Accepts multipart/form-data with image file or JSON body with base64 data URI. Returns public image URL for use in milestone status updates.

```
{
  "status": "success",
  "message": "Proof photo uploaded successfully",
  "url": "https://api.bemsfarms.com/uploads/proofs/POD_178992819_3c8f.jpg",
  "filename": "POD_178992819_3c8f.jpg",
  "size": 184520,
  "mimetype": "image/jpeg"
}
```

5. Driver Wallet, Bank Verification & Withdrawal System

Provides live wallet balance, Monnify Dedicated Virtual Accounts (DVA NUBAN), Nigerian commercial bank directory, live NIP account name resolution, and withdrawal processing.

5.1 Get Driver Wallet & Earnings (GET /api/driver/earnings)

```
{
  "wallet": {
    "total_earned": 28500.00,
    "total_paid": 12000.00,
    "pending_payouts": 0.00,
    "available_balance": 16500.00,
    "commission_per_delivery": 1750.00,
    "dedicated_virtual_account": {
      "account_number": "8559127267",
      "bank_name": "Monnify / Wema Bank",
      "account_name": "BEMS - VICTOR KALU"
    },
    "withdrawal_bank": {
      "bank_name": "GTBank",
      "account_number": "0123456789",
      "account_name": "VICTOR KALU"
    }
  },
  "zone_rates": [
    { "zone_id": "ZONE001", "zone_name": "Zone 1 - Maitama & Asokoro", "driver_earning_fee": 700.00, "commission_percent": 70, "estimated_eta": "20 - 35 mins" },
    { "zone_id": "ZONE002", "zone_name": "Zone 2 - Garki, Wuse, Central Area", "driver_earning_fee": 1750.00, "commission_percent": 70, "estimated_eta": "25 - 45 mins" },
    { "zone_id": "ZONE003", "zone_name": "Zone 3 - Utako, Jabi, Mabushi", "driver_earning_fee": 2450.00, "commission_percent": 70, "estimated_eta": "30 - 50 mins" },
    { "zone_id": "ZONE004", "zone_name": "Zone 4 - Gwarinpa, Kubwa", "driver_earning_fee": 3500.00, "commission_percent": 70, "estimated_eta": "40 - 65 mins" },
    { "zone_id": "ZONE005", "zone_name": "Zone 5 - Lugbe, Airport Road", "driver_earning_fee": 5600.00, "commission_percent": 70, "estimated_eta": "45 - 75 mins" },
    { "zone_id": "ZONE006", "zone_name": "Zone 6 - Interstate Express", "driver_earning_fee": 24500.00, "commission_percent": 70, "estimated_eta": "Same Day" }
  ]
}
```

5.2 List Nigerian Commercial Banks & Fintechs (GET /api/driver/banks)

Returns official directory of 33+ Nigerian banks (Access, GTBank, Zenith, OPay, PalmPay, Kuda, Moniepoint, Wema, etc.) with codes for populating bank picker modals.

5.3 Resolve & Verify Bank Account Name (POST /api/driver/bank/resolve)

Validates 10-digit NUBAN account number against the bank code via NIP name enquiry. The app calls this as soon as 10 digits are typed to display the verified account holder name.

Request Payload	Success Response (200 OK)
<pre>{ "account_number": "0123456789", "bank_code": "058", "bank_name": "Guaranty Trust Bank (GTBank)" }</pre>	<pre>{ "status": "success", "account_number": "0123456789", "bank_code": "058", "bank_name": "Guaranty Trust Bank (GTBank)", "account_name": "VICTOR KALU", "is_valid": true, "message": "Account name successfully resolved: VICTOR KALU" }</pre>

5.4 Request Withdrawal (`POST /api/driver/withdraw`)

Request Payload	Success Response (201 Created)
<pre>{ "amount": 10000.00, "bank_name": "Guaranty Trust Bank (GTBank)", "bank_code": "058", "account_number": "0123456789", "account_name": "VICTOR KALU", "notes": "Weekly earnings withdrawal" }</pre>	<pre>{ "message": "Withdrawal request submitted successfully.", "payout": { "payout_ref": "PAY-MNFY-99201", "amount": 10000.00, "bank_name": "Guaranty Trust Bank (GTBank)", "account_number": "0123456789", "account_name": "VICTOR KALU", "status": "pending" }, "remaining_available_balance": 6500.00 }</pre>

6. Push Notifications & In-App Feed System

6.1 Register Device Push Token (`POST /api/driver/device-token`)

Registers the mobile device's Expo or Firebase (FCM) push token so the server can push delivery alerts when the app is in the background.

Request Payload	Success Response (200 OK)
<pre>{ "token": "ExponentPushToken[xxxxxxxxxxxxxxxxxxxx]", "platform": "expo", "device_info": { "os": "iOS 17.4", "model": "iPhone 15 Pro" } }</pre>	<pre>{ "status": "success", "message": "Device push token registered successfully" }</pre>

6.2 Fetch In-App Notifications (`GET /api/driver/notifications`)

<pre>{ "status": "success", "total": 14, "unread_count": 2, "notifications": [{ "id": 101, "title": "New Dispatch Assigned", "body": "Order #BEMS-9901 is ready for pickup at Hub.", "type": "dispatch", "is_read": false, "created_at": "2026-09-21T10:05:00Z" }, { "id": 98, "title": "Withdrawal Approved", "body": "₦10,000.00 has been paid to your GTBank account.", "type": "payout", "is_read": true, "created_at": "2026-09-20T16:20:00Z" }] }</pre>

7. Mid-Trip Incident & Problem Reporting

7.1 Report Delivery Problem (`POST /api/driver/deliveries/:orderId/report-issue`)

Allows drivers to open an operational problem ticket directly with dispatch when encountering issues on the road.

Request Payload	Success Response (201 Created)
<pre>{ "issue_type": "customer_unreachable", "description": "Called customer 4 times, phone switched off at gate.", "latitude": 9.0765, "longitude": 7.3986, "photo_urls": ["https://api.bemsfarms.com/uploads/proofs/gate_photo.jpg"] }</pre>	<pre>{ "status": "success", "message": "Incident report submitted to dispatch operations", "incident": { "incident_ref": "INC-MNFY-7712", "issue_type": "customer_unreachable", "status": "open" } }</pre>

8. Emergency SOS / Safety Panic System

8.1 Trigger Emergency SOS (`POST /api/driver/emergency`)

High-priority safety endpoint that instantly alerts the Operations Control Room with the driver's exact coordinates and vehicle details.

Request Payload	Success Response (201 Created)
<pre>{ "latitude": 9.0765, "longitude": 7.3986, "address": "Maitama Junction, Near Transcorp Hilton", "battery_level": 42, "emergency_type": "vehicle_accident", "notes": "Rear-ended by another vehicle, need immediate assistance." }</pre>	<pre>{ "status": "success", "message": "Emergency SOS alert activated. Dispatch Operations team has been notified immediately.", "emergency": { "emergency_ref": "SOS-MNFY-9910", "status": "active" } }</pre>

9. Driver Performance Scorecard & Customer Ratings

9.1 Get Performance Scorecard (`GET /api/driver/stats`)

```
{
  "status": "success",
  "scorecard": {
    "customer_rating": 4.92,
    "total_ratings_count": 84,
    "acceptance_rate": 97.5,
    "on_time_delivery_rate": 98.2,
    "total_km_driven": 412.80
  },
  "deliveries_summary": {
    "completed_today": 6,
    "completed_this_week": 28,
    "completed_this_month": 112,
    "total_completed": 112
  },
  "badges": [
    { "id": "top_rated", "name": "Top Rated Star", "earned": true, "icon": "★" },
    { "id": "century_rider", "name": "100 Deliveries Club", "earned": true, "icon": "🏆" },
    { "id": "speedy_courier", "name": "Fast Dispatcher", "earned": true, "icon": "🚀" }
  ]
}
```

10. Zone-Based Compensation Rate Schedule (Zones 1–6)

Zone ID	Zone Coverage Area	Customer Delivery Fee	Driver Drop Payout (70%)	Estimated ETA
ZONE001	Zone 1 - Maitama & Asokoro	₦1,000.00	₦700.00	20 - 35 mins
ZONE002	Zone 2 - Garki, Wuse, Central Area	₦2,500.00	₦1,750.00	25 - 45 mins
ZONE003	Zone 3 - Utako, Jabi, Mabushi	₦3,500.00	₦2,450.00	30 - 50 mins
ZONE004	Zone 4 - Gwarinpa, Kubwa Expressway	₦5,000.00	₦3,500.00	40 - 65 mins
ZONE005	Zone 5 - Lugbe, Airport Road	₦8,000.00	₦5,600.00	45 - 75 mins
ZONE006	Zone 6 - Interstate / Regional Express	₦35,000.00	₦24,500.00	Same Day / Next Day

11. Mobile Client Engineering Checklist

- **1. Secure Storage:** Store the JWT token securely using SecureStore (Expo) / EncryptedSharedPreferences (Android) / Keychain (iOS).
- **2. Push Token Registration:** Call POST /api/driver/device-token upon app launch and notification permission grant.
- **3. Background GPS Streaming:** When duty state is online (is_available: true), stream GPS coordinates to POST /api/driver/location every 10–15s.
- **4. Navigation Deep Linking:** Deep-link customer coordinates to Google Maps (geo:\${lat},\${lng}?q=\${lat},\${lng}) or Apple Maps (maps://?daddr=\${lat},\${lng}).
- **5. Direct Customer Contact:** Provide 1-tap dialer (tel:\${phone}) and WhatsApp quick chat (https://wa.me/234\${phone}).
- **6. Live Bank Validation:** Call POST /api/driver/bank/resolve dynamically upon entering 10 digits in the withdrawal modal to preview verified account name.
- **7. POD Camera Capture:** Upload handover pictures via POST /api/driver/upload/proof prior to calling PATCH /status with status `delivered`.
- **8. 1-Tap SOS Button:** Provide a persistent SOS panic button on active delivery screens linking to POST /api/driver/emergency.